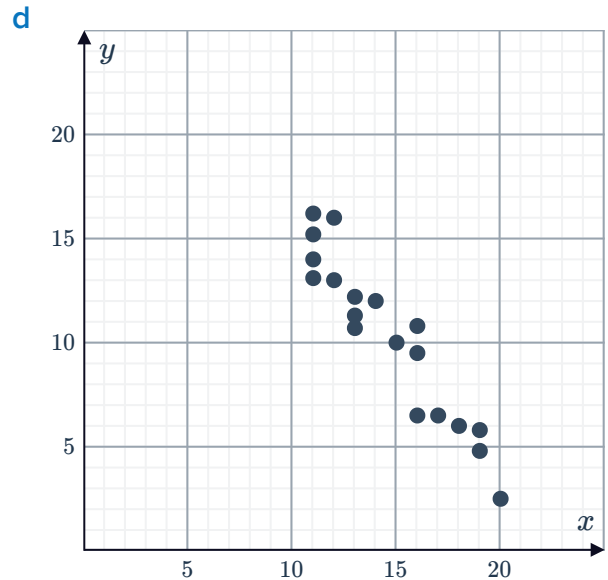
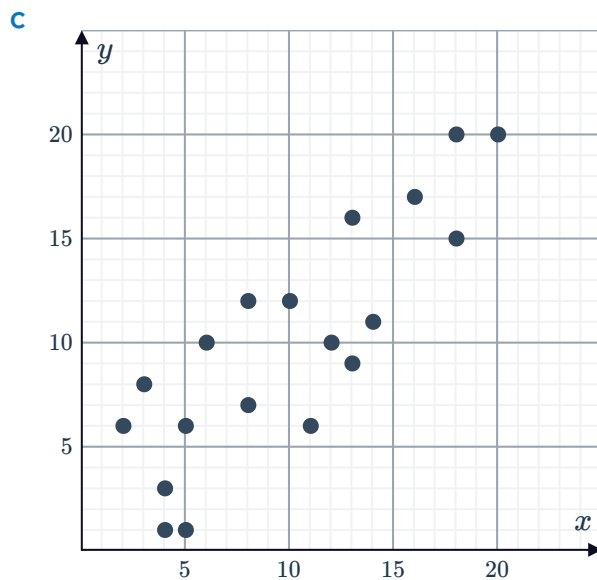
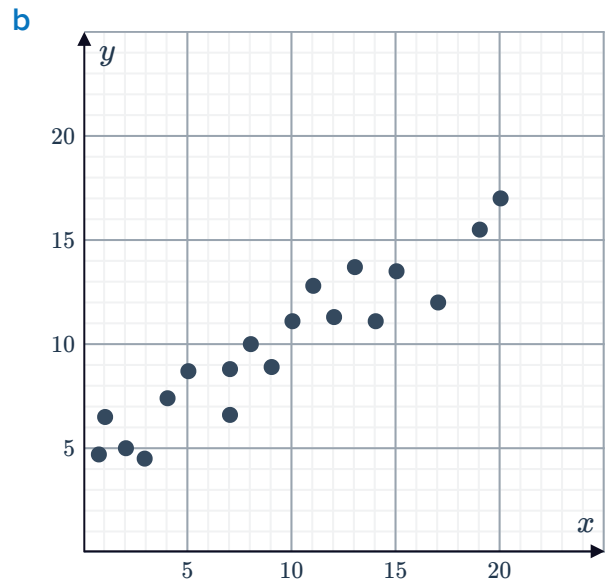
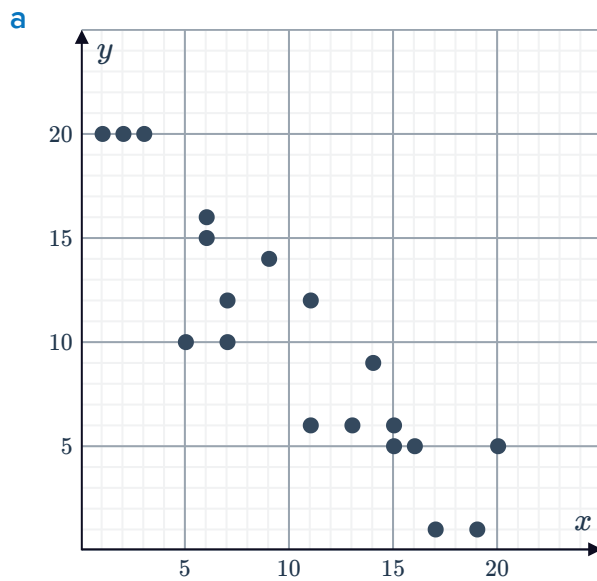


Line of best fit

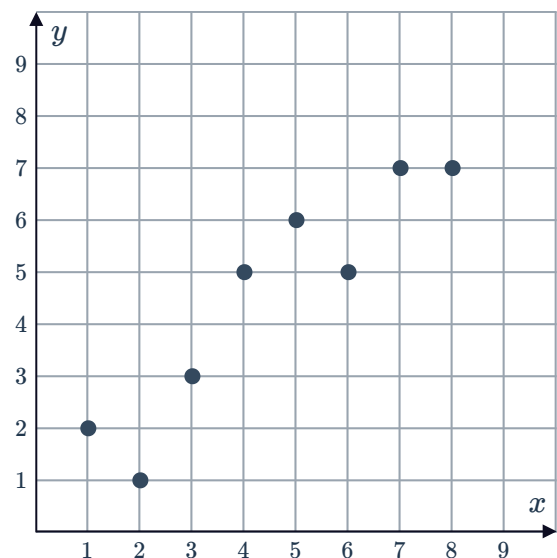


1 Draw an approximate line of best fit by hand for each of the the scatter plots below:



2 The following scatter plot shows the data for two variables, x and y :

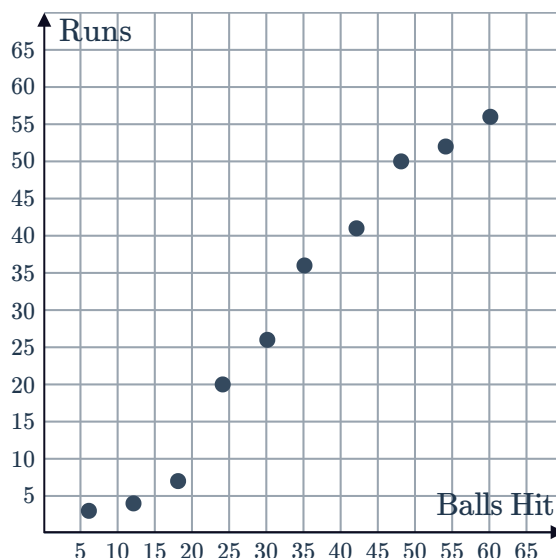
- Sketch the line of best fit for this data.
- Use your line of best fit to estimate the value of y when:
 - $x = 4.5$
 - $x = 9$



- 3 The following scatter plot graphs data for the number of balls hit and the number of runs scored by a batsman:

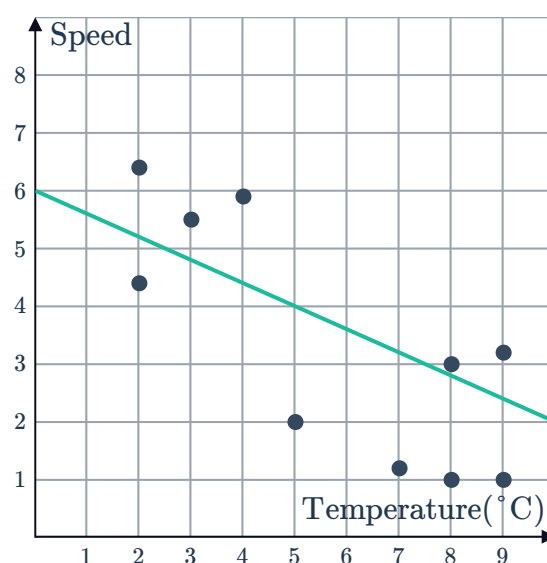


- Sketch the line of best fit for this data.
- Use the line of best fit to estimate the number of runs scored by the batsman after hitting:
 - 27 balls
 - 66 balls
- Is the relationship between the two variables positive or negative?



- 4 The average monthly temperature and the average wind speed, in knots, in a particular location was plotted over several months. The graph shows the points for each month's data and their line of best fit:

Use the line of best fit to approximate the wind speed on a day when the temperature is 5°C .



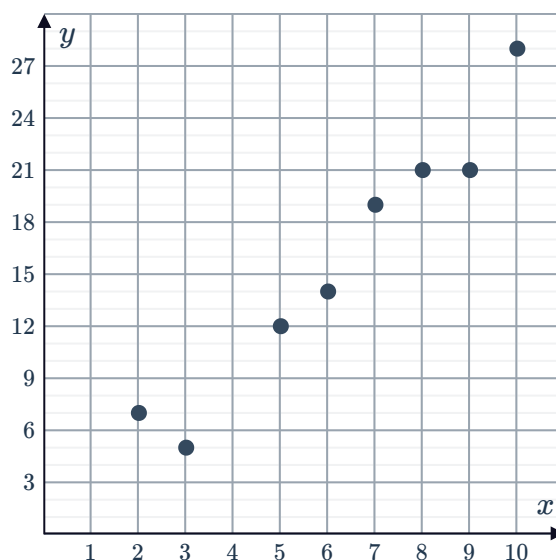
Equation of a line of best fit

- 5 Consider the following scatter plot:

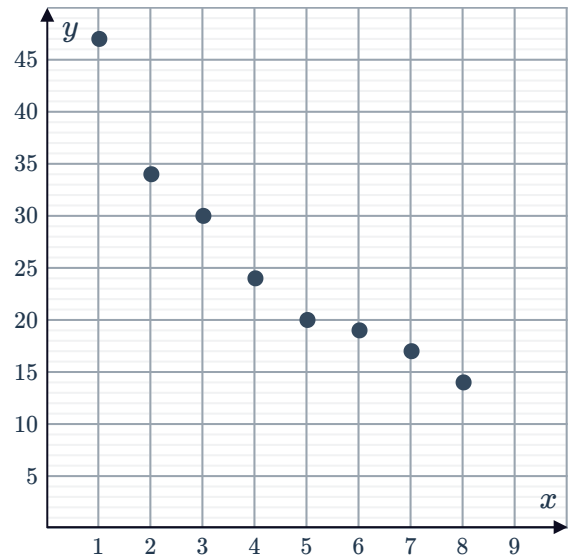
- Is the relationship between the x and y variables positive or negative?
- Sketch the line of best fit for this data.
- Which of the following could be the equation for the line of best fit:

A $y = 2 - 3x$ **B** $y = 3x + 2$

C $y = -3x - 2$ **D** $y = 3x - 2$



6 Consider the following scatter plot:



- a Is the relationship between the x and y variables positive or negative?
- b Which of the following could be the equation for the line of best fit:

A $y = -4x - 4$ **B** $y = 44 + 4x$

C $y = -4x + 44$ **D** $y = 4x - 4$

7 Use technology to find the line of best fit for the sets of data below. Write the equation with the coefficient and constant term to the nearest two decimal places.

a

x	24	37	19	31	32	22	14	30	23	40
y	-7	-8	-3	-6	-9	-8	-2	-8	-8	-12

b

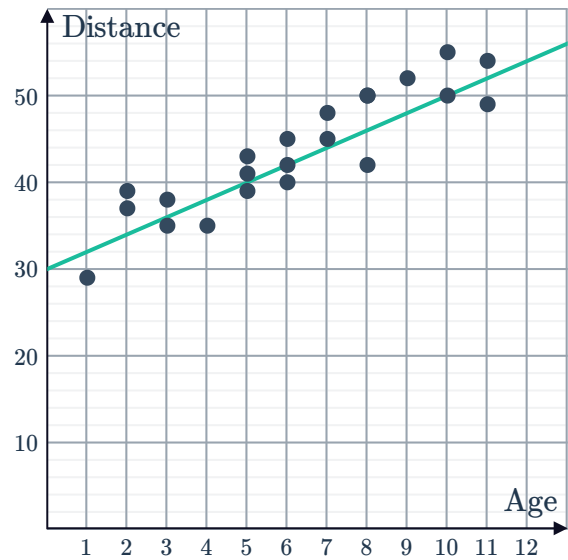
x	7	18	16	9	16	19	8	8	12	7
y	16.29	3.56	7.25	12.11	8.25	3.26	12.5	12.5	10.33	14.29

c

x	44	39	41	50	45	55	48	54	44	43
y	2.29	1.57	3.86	3.14	4.43	4.86	3.86	4.71	4.29	4.14

- 8 Several cars underwent a brake test and their age, x (in years), was measured against their stopping distance, y (in metres). The scatter plot shows the results and a line of best fit that approximates the positive correlation:

- According to the line, what is the stopping distance of a car that is 6 years old?
- According to the line, what is the stopping distance of a car that is 10 years old?
- Using the information found above, determine the gradient of the line of best fit.
- State the value of the vertical intercept of the line.
- Use the line of best fit found to estimate the stopping distance of a car that is 4.5 years old.



- 9 The distance of several locations from the equator and their temperature on a particular day is measured. The values are presented on the following scatter plot:

- Determine whether the following could be the equation of the line relating distance (x) and temperature (y):

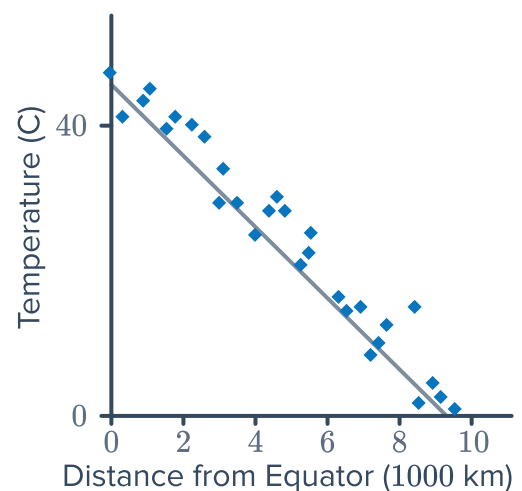
A $y = -0.005x + 49$

B $y = -0.005x - 49$

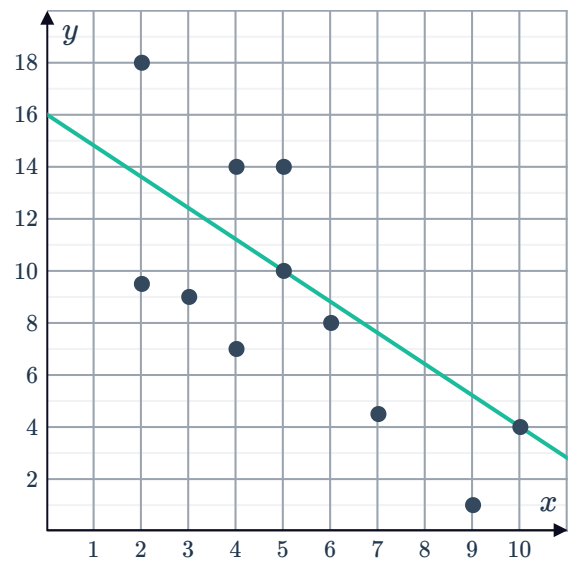
C $y = 0.005x + 49$

D $y = 0.005x - 49$

- Estimate the distance from the equator, x , if the temperature is 30.59°C .



- 10 Find the equation of the line of best fit on the scatter plot shown:



- 11 Consider the scatter plot shown:

- a Find the equation of the line of best fit.
- b Use the line of best fit to approximate the value of y for $x = 6.9$.

